

ConvertAudioEngine-DDP51.ps1 v3.1.2

Converts all audio tracks (7.1 → downmix, 5.1 → re-encode, passthrough where applicable). Only stereo and commentary tracks are removed.

- Downmixes **7.1 audio** to **DDP 5.1 (1152k)** via ITU-R BS.775 pan matrix
- Re-encodes **5.1 audio** to **DDP 5.1 (768k)**; DTS-HD MA/HRA re-encodes at **1024k**
- Passthrough for **EAC3 Atmos**, **EAC3 5.1**, and **TrueHD 5.1**
- All **2.0 stereo tracks are removed** — stereo output is handled by ConvertAudioEngine-Stereo
- Maintains perfect sync on long-duration files
- Normalizes malformed channel layouts
- Removes 2-channel commentary tracks

Requirements

PowerShell 7.6 - [Click here to Download PS 7.6 \(Windows x64 .msi\)](#)

FFmpeg 8.1 - <https://ffmpeg.org/download.html>

Quick Start (Windows 11)

1. Place **ConvertAudioEngine-DDP51.ps1** , **ffmpeg.exe** , and **ffprobe.exe** in the same folder.
2. Run the script:

```
pwsh -ExecutionPolicy Bypass -File .\ConvertAudioEngine-DDP51.ps1 ".\YourMovie.mkv"
```

The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

macOS / Linux

1. Place the script, **ffmpeg** , and **ffprobe** in the same folder.
2. Make the binaries executable (first time only):

```
chmod +x ffmpeg ffprobe
```

3. Run:

```
pwsh -File ./ConvertAudioEngine-DDP51.ps1 "./YourMovie.mkv"
```

1. To Adjust Bitrate (line 70-74)

```
You can safely adjust these numbers, within ''....k''
Common EAC3 5.1 bitrates: 384, 448, 512, 640, 768
Higher pipeline bitrates: 1024, 1152, 1280, 1408, 1536

$BitRateConfig = @{
    Downmix51    = '1152k'    # << Change your Bitrate
    ReEncode51   = '768k'     # << Change your Bitrate
    EncodeDTSHD  = '1024k'    # << Change your Bitrate
```

Remember to save the file after making changes.

All three values can be adjusted safely — changes apply automatically throughout the script.

Key	Default	Used for
Downmix51	1152k	All 7.1 sources downmixed to 5.1
ReEncode51	768k	5.1 sources re-encoded to DD+
EncodeDTSHD	1024k	DTS-HD MA and HRA sources (5.1 and 7.1)

Supported Audio Codecs

The engine supports and processes the following codec families:

Codec Family	Formats Included
AAC	AAC LC, HE-AAC, AAC 5.1, AAC 7.1
EAC3 / Atmos	EAC3, EAC3-JOC (Atmos)
TrueHD	TrueHD 5.1, TrueHD 7.1, TrueHD+Atmos
DTS	DTS Core, DTS-HD MA, DTS-HD HRA
PCM	pcm_s16le, pcm_s24le, pcm_f32le, pcm_f32be
FLAC	All FLAC channel layouts

2. FFmpeg Global Options

These are global flags to ensure FFmpeg handles large, high-resolution files without data corruption or sync drift.

- **-y** — overwrites the output file without asking.
- **-loglevel error** — suppresses all output except actual errors.
- **-analyzeduration 200M / -probesize 200M** — widens the initial probe buffer to 200 MB; high-bitrate audio like TrueHD+Atmos or AAC 7.1 needs a deep probe to fully resolve the channel layout, since standard 5–20 MB probes often misidentify these tracks.
- **-err_detect ignore_err** — bypasses minor bitstream errors so one corrupted audio frame (common in digital rips) doesn't abort a multi-hour conversion.
- **-drc_scale 0** — disables Dynamic Range Compression, preserving the source's full dynamic range rather than flattening loud peaks.

2.5 ScanThrottle + ThreadCount

Sets how many CPU threads the encoder uses (line 47).

```
$ThreadCount = 8      # User-adjustable (4-16).
```

Limits how many peak-scan tasks run at the same time (line 49).

```
$ScanThrottle = [Math]::Max(1, [int]([Environment]::ProcessorCount /  
$ThreadCount))  
# Recommended overrides: 1, 2, 3, or 4
```

What ScanThrottle does:

- Limits parallel SPN scans to avoid CPU overload.
- Keeps the system responsive during peak detection.
- Does not affect encoding speed.
- SSD/NVMe: safe to override up to 4. Spinning disk: keep at 1.

3. The "De-Sync" Safeguard

These flags are applied after the input to maintain perfect alignment on long-duration files.

- `-avoid_negative_ts make_zero` — clamps negative initial PTS (common in source files) to zero, preventing audio lag or lead.
- `-max_muxing_queue_size 14000` (*1024–100000 recommended range*) — raises muxing buffer headroom. Downmixing and re-encoding (AAC 7.1, TrueHD 7.1, ...) can run slower than the muxer, so 14000 provides enough headroom to prevent queue overflows on all movie lengths.
- `-map 0:v? -c:v copy` — maps all video tracks in stream-copy mode for zero quality loss at maximum speed.
- `-map_metadata 0 -map_chapters 0` — copies global metadata and chapter markers (title, year, scene markers) from the source.

3.2 Safe Peak Normalizer (SPN)

- **What:** Safe Peak Normalizer (SPN-TP) measures each non-passthrough track's true peak before encoding and only steps in when the measured true peak exceeds the ceiling (above -1.0 dBTP).
- **Why:** Some movies arrive with peaks already above 0 dBFS. When that happens, even simple mixing of the channels can push those peaks higher and cause distortion in the final encode. SPN catches this early so the issue never reaches the rest of the audio processing.
- **How:** Safe Peak Normalizer runs right after the script decides each track's action (Encode, Downmix, etc.). If the scan shows the true peak is above the ceiling, a static, attenuation-only `volume trim` (always ≤ 0 dB — never a boost) is inserted into that track's filter chain: after the pan on the downmix path, ahead of the encoder on the re-encode path. An `alimiter` backstop sits just below at ~ -0.47 dBFS.
- **Clean Sources:** Nothing is changed — the original loudness and dynamics stay exactly the same.
- **Passthrough tracks:** — Peak detection is skipped entirely. Copied streams cannot have filters applied.
- **Dirty Sources:** SPN fixes unsafe peaks without making the movie sound compressed or “over-normalized.”
- **Result:** More consistent, safer audio across all movies, keeping clean sources untouched.

True-Peak Ceiling (line 53)

This is the value SPN-TP uses to decide when to step in.

```
[double]$TruePeakCeilingDB = -1.0 # SPN-TP: static trim engages when measured true peak exceeds this (dBTP)
```

- Recommended: **-1.0 dBTP**.
- More negative (e.g. -2.0) trims more tracks and leaves extra headroom — more conservative.
- Closer to 0 (e.g. -0.5) trims fewer tracks — only the hottest sources.
- The trim is attenuation-only and clamped to ≤ 0 dB; SPN-TP never boosts.

Setting	Value
True-peak ceiling (\$TruePeakCeilingDB)	-1.0 dBTP
Trim type	static volume , attenuation-only (≤ 0 dB)
loudnorm	measurement only (print_format=json) — no normalization applied
alimiter backstop	limit=0.948 (~-0.47 dBFS)

Three-Stage SPN Processing

The audio engine processes each audio track in three phases, with SPN peak scans running in parallel during Phase B to keep multi-track processing fast.

```
3-phase SPN Architecture (Phase A > Phase B > Phase C)
Input MKV >> Probe
    > Phase A (rule matching + malformed-layout guards + commentary removal)
    > Phase B (parallel SPN peak scans)
    > Phase C (sequential merge + true-peak safety resolution)
    > FFmpeg command builder
    >> Output MKV
```

Phase A, Sequential rule matching

Every audio stream is probed and matched against the rule table. Malformed-layout guards and commentary detection run here. 2.0 tracks are marked Removed. Surviving tracks receive a pending action (Passthrough, Downmix, Encode).

Phase B, Parallel SPN-TP true-peak scans

All non-Removed, non-Passthrough tracks are scanned in parallel for true peak using **loudnorm**

(`print_format=json`). Downmix tracks — and `>6ch` Encode tracks (DTS-HD MA/HRA 7.1), which the builder folds to 5.1 through that same chain — are measured through the pan chain the encoder uses, so the reading reflects the 5.1 output peak rather than the 7.1 source.

Phase C, Sequential merge

Peak results are merged back into the track list. The attenuation-only true-peak safety gain (`NeedsPeakSafety`, `min(0, ceiling - measured_tp)`) is resolved from the scan data. The final track list is handed to the FFmpeg command builder.

4. Encoding & Downmix Execution

These parameters determine how the audio is shaped and processed during encoding.

- `-filter:a:$i $panFilter` — applies the 7.1→5.1 pan matrix on the Downmix path; see § 6 for the matrix and its rationale.
- `-c:a:$i eac3` / `-b:a:$i $t.Bitrate` — EAC3 (DD+) at the bitrate from the rule table (§ 5) or `$BitRateConfig` .
- `-ac 6` — forces exactly 6 channels on the ≤ 6 ch branch of the 5.1 re-encode path, guaranteeing correct channel count even on a non-standard source layout. Not needed on the `>6ch` branch (DTS-HD MA/HRA 7.1), where the pan filter already outputs 5.1.
- `-ar 48000` — normalizes the output sample rate to 48 kHz on both encode paths (Downmix and 5.1 re-encode). DD+ delivery is 48 kHz; this brings high-rate lossless sources (96/192 kHz PCM/FLAC) down to spec. Not applied to passthrough streams.
- `-dialnorm -31` — sets Dialogue Normalization to the reference "off" position, preventing the playback device from applying its own volume leveling on top of the mix's intended loudness.
- `-cutoff 20000` — sets the low-pass filter to 20 kHz, preserving high-frequency content that some encoders discard by default. Applies to the Downmix and 5.1 re-encode paths only — not passthrough.
- `-metadata:s:a:$i "title=..."` — labels the track (e.g., "DD+ 5.1 Downmix") so it's identifiable in a media player's audio menu.

5. Audio Rules & Priority

The audio-engine uses a **Priority Mapping (0-100)** to evaluate tracks based on quality and codec type.

EAC3 / Atmos Rules

Priority	Condition	Action	Output
100	Atmos (JOC)	Passthrough	Original EAC3 Atmos — preserves object metadata
99	EAC3 7.1	Downmix	DDP 5.1 @ 1152k — maps 8 channels to 6 using pan filter
98	EAC3 5.1	Passthrough	Original EAC3 5.1

97	EAC3 2.0	Removed	Delegated to ConvertAudioEngine-Stereo
----	----------	---------	--

v3.1.1 — Profile-tier matching: Tier promotion now trusts only the `ffprobe` `profile` field. Container title/handler metadata is consulted solely where a rule opts in via `TrustTitleProfile` — currently the EAC3 Atmos rule alone (Priority 100), because `ffprobe` under-reports JOC/Atmos in the `profile` field. This blocks a mislabeled track (e.g. a DTS 5.1 core titled "TrueHD Atmos 7.1") from being promoted into a higher encode tier through a title/handler substring match. The Atmos match is word-boundaried: `\bJOC\b|\bAtmos\b`.

TrueHD Rules

Priority	Channels	Action	Output
81	5.1	Passthrough	Original TrueHD 5.1
80	7.1	Downmix	DDP 5.1 @ 1152k

AAC Rules

Priority	Channels	Action	Output
92	7.1	Downmix	DDP 5.1 @ 1152k
91	5.1	Encode	DDP 5.1 @ 768k
90	2.0	Removed	Delegated to ConvertAudioEngine-Stereo

DTS Rules

Priority	Variant	Action	Output
73	DTS-HD MA/HRA (any ch > 2)	Encode	DDP 5.1 @ 1024k
72	DTS Core (any ch > 2)	Encode	DDP 5.1 @ 768k
71	DTS-HD 2.0	Removed	Delegated to ConvertAudioEngine-Stereo
70	DTS 2.0	Removed	Delegated to ConvertAudioEngine-Stereo

PCM / FLAC Rules

Priority	Channels	Action	Output
62	7.1	Downmix	DDP 5.1 @ 1152k

61	5.1	Encode	DDP 5.1 @ 768k
60	2.0	Removed	Delegated to ConvertAudioEngine-Stereo

6. Spatial Audio: The Pan Filter Matrix

When 7.1 audio is reduced to 5.1, standard downmixing often loses spatial detail from the side channels.

The Formula

```
aformat=channel_layouts=7.1,pan=5.1|FL=FL|FR=FR|FC=FC|LFE=LFE|BL=BL+0.707SL/BR=BR+0.707SR
alimiter=limit=0.948:attack=5:release=50:level=false:latency=1
```

Layout Locking (aformat)

What: Forces the decoder to output a strict 8-channel 7.1 layout *before* the Pan Filter.

Why: Some codecs (TrueHD, AAC, EAC3) may decode into non-standard 7.1 variants.

Locking the layout ensures:

- SL really is the Side Left channel
- BL really is the Back Left channel
- No channel swapping
- No silent corruption of the downmix

The Math (0.707)

What: Applies a -3 dB coefficient to the Side Surrounds (SL/SR).

Why: Prevents the side channels from overpowering the rear soundstage and avoids digital clipping.

Limiter (0.948) — For All 7.1 Codecs

alimiter=limit=0.948:attack=5:release=50:level=false:latency=1

Why: All 7.1 codecs can produce true peak overshoots after decoding:

- AAC 7.1 > AAC can create peaks slightly above 0 dB after decoding.
- TrueHD 7.1 > Combining similar channels can briefly push the volume higher.
- DTS-HD MA > Decoder can rebuild peaks that go a bit above the original level.
- PCM/FLAC > Short volume spikes can appear when channels are mixed together.
- EAC3 7.1 > Channels are merged into fewer speakers during downmixing.

Limiter ceiling of 0.948 provides about 0.5 dB of true-peak headroom, preventing:

- 1.) +0.0x dB overshoots
- 2.) clipping

latency=1 allows the limiter a 1-sample look-ahead window to catch post-downmix peaks without adding audible delay.

Important:

- If the source audio already contains true-peak or inter-sample peaks (“clips”), these will also appear in **analysis** of the downmix.
- The limiter prevents new clipping during downmixing and encoding, but **it does not remove overshoots inherent in the source mix.**

7. Commentary Filtering

Only 2-channel commentary tracks are removed to save space while preserving primary content.

- **Detection Methods:** Title matching, keyword detection, and case-insensitive search.
- **Keywords:** commentary, director, producer, writer, cast, behind, bonus, alt, interview.
- **Why 5.1 Commentary is NOT removed:**
 - It is extremely rare in retail media.
 - It is often hard to tell apart from legitimate "Alternate Mixes" or "Home Theater Optimized" tracks.
 - Removing it carries a high risk of "False Positives" (deleting real movie content).

8. Malformed Layout Guards

The script corrects illegal 7-channel layouts before rule matching runs:

- **AAC 7ch** → 6ch (treated as 5.1)
- **EAC3 7ch** → 6ch (treated as 5.1)
- **TrueHD 7ch** → 8ch (treated as 7.1)
- **PCM/FLAC 7ch** → 8ch (treated as 7.1)
- **AC3 7ch** → Removed (no valid AC3 7-channel layout exists)

Fallback Rules

If no specific rule matches, the following logic is applied:

Channels	Fallback Action	Bitrate
2 channels or fewer	Removed	(delegated to ConvertAudioEngine-Stereo)
3 to 6 channels	Encode	768k
more than 6 channels (7ch+)	Downmix	1152k

9. Summary Report

At === AUDIO PROCESSING SUMMARY ===, the script generates a detailed table with the following columns:

Column	Description
Index	The original stream number (0, 1, 2, 3,).
Codec	The source format (e.g., TrueHD, DTS).
Channels	The detected number of channels (e.g., 7.1, 5.1).
Action	Green : Passthrough, Yellow : Downmix, Blue : Encode, Red : Removed.
Output	The final codec and bitrate configuration.
Priority	The assigned priority level (0–100). Removed tracks show  .
Rule	the logical rule that was matched during processing, such as "TrueHD 7.1 Rule"

10. 2.0 Audio Encoding — Why It's Disabled

All 2.0 rules are present in the engine but set to `Action = "Removed"` by default. This is intentional — stereo output is delegated to **ConvertAudioEngine-Stereo**, and **the engine has no native stereo encode path**. That second point is what makes re-enabling a two-step change, not a one-step one.

⚠ **Changing the rule alone upmixes 2.0 → 5.1.** Any `Action = "Encode"` track is routed through the 5.1 re-encode branch of `Build-FFmpegCommand`, which unconditionally applies `-ac 6`. Flip a 2.0 rule to `Encode` and nothing else, and the source is **upmixed to DD+ 5.1** at your chosen bitrate — *not* emitted as a 2.0 track. For most uses that is not what you want; leaving 2.0 delegated to `ConvertAudioEngine-Stereo` is the recommended default.

11. Non-Critical FFmpeg Warnings (Safe to Ignore)

During processing, FFmpeg may display one or more of the following messages.

These warnings are **normal**, and do not affect audio quality, sync, or re-encode.

“quant_step_size larger than huff_lsbs”

- **Cause:** Printed by the TrueHD decoder during decoding (not probe).
- **Impact:** Safe to ignore. Does not affect decoding, channel layout, or the Pass+Copy path.

“Codec AVOption drc_scale ... has not been used for any stream”

- **Cause:** (`-drc_scale`) applies only to AC-3/EAC3 decoders. FFmpeg prints this when the input track is not AC-3 (e.g., TrueHD, DTS, FLAC).
- **Impact:** Safe to ignore. The script disables DRC intentionally for consistent output.

“(Subtitle: hdmv_pgs_subtitle) unspecified size”

- **Cause:** PGS (Blu-ray) subtitles do not declare width/height in the container. FFmpeg warns until the first subtitle packet is decoded.
- **Impact:** Safe to ignore. Subtitles are copied untouched (`-c:s copy`).

“Assuming an incorrectly encoded 7.1 channel layout...instead of 7.1(wide)”

- **Cause:** AAC has two 7.1 layouts. Most encoders use the common (non-spec) layout. FFmpeg prints this when the file does not use the strict 7.1(wide) AAC layout.
- **Impact:** Safe to ignore. Script corrects and pins layout using (`aformat=channel_layouts=7.1`)

Disclaimer

- Always test on a few files before batch-processing your library.
- Keep backups of original media.
- Results may vary depending on source quality and hardware.